

Microeconomics II – Class 8

Axioms of revealed preference



Class summary

- * How to check the Weak Axiom of Revealed Preference
- * Strong Axiom of Revealed Preference
- * How to check the Strong Axiom of Revealed Preference
- * Index numbers
 - * Laspeyres and Paasche quantity and price indexes
 - * Expenditure index
 - * Relationship among Laspeyres and Paasche quantity and price indexes, the expenditure index, revealed preferences and welfare.
- * Readings: Varian (chap. 7)



How to check the Weak Axiom of Revealed Preference (WARP)

- * Consider the following 3 observations (A, B e C)

Observation	p1	p2	x1	x2
A	1	2	1	2
B	2	1	2	1
C	1	1	2	2

- * Is this consumer's behavior consistent with the economic theory of an utility-maximizing consumer?



How to check the Weak Axiom of Revealed Preference (WARP)

- * 1st step: Construct a matrix with the necessary expenditure to buy each consumption bundle for each given price level

Observation	p_1	p_2	x_1	x_2
A	1	2	1	2
B	2	1	2	1
C	1	1	2	2

Expenditure table		Consumption bundles		
		A	B	C
Prices	A			
	B			
	C			



How to check the Weak Axiom of Revealed Preference (WARP)

* Expenditure incurred:

$$p_1^i \times x_1^i + p_2^i \times x_2^i$$

Observation	p1	p2	x1	x2
A	1	2	1	2
B	2	1	2	1
C	1	1	2	2

Expenditure table		Consumption bundles		
		A	B	C
Prices	A	5		
	B		5	
	C			4



How to check the Weak Axiom of Revealed Preference (WARP)

- * Expenditure required to buy an alternative bundle:

$$p_1^i \times x_1^j + p_2^i \times x_2^j$$

Observation	p1	p2	x1	x2
A	1	2	1	2
B	2	1	2	1
C	1	1	2	2

Expenditure table		Consumption bundles		
		A	B	C
Prices	A	5	4	6
	B	4	5	6
	C	3	3	4



How to check the Weak Axiom of Revealed Preference (WARP)

- * 2nd step: For each price level (each row), mark (with an asterisk, for example) the bundles which despite being affordable were not chosen/bought.

Expenditure table		Consumption bundles [$X \equiv (x_1, x_2)$]		
		A	B	C
Prices [$P \equiv (p_1, p_2)$]	A	5	4*	6
	B	4*	5	6
	C	3*	3*	4



How to check the Weak Axiom of Revealed Preference (WARP)

- * 3rd step: Look for situations where both cells (i,j) and (j,i) have an asterisk.

Tabela de despesa		Cabazes $[X \equiv (x_1, x_2)]$		
		A	B	C
Preços $[P \equiv (p_1, p_2)]$	A	5	4*	6
	B	4*	5	6
	C	3*	3*	4

$A \succ B \wedge B \succ A \Rightarrow$ The Weak Axiom of Revealed Preferences (WARP) does not hold



Strong Axiom of Revealed Preference

- * Similar to the weak axiom, but including the indirect preference revelation by transitivity

If $(x_1, x_2) \succ (y_1, y_2)$ (directly or indirectly)
and $(x_1, x_2) \neq (y_1, y_2)$ then we must not have
 $(y_1, y_2) \succ (x_1, x_2)$ (directly or indirectly)

- * The Strong Axiom of Revealed Preference (SARP) is a necessary and sufficient condition for consistency between the consumer's behavior and the economic theory of an optimizing consumer



How to check the Strong Axiom of Revealed Preference (SARP)

- * The 1st and 2nd steps are similar to the procedure adopted to check WARP
- * 3rd step: Use transitivity to look for indirect revelation of preferences
- * New example:

Expenditure table		Consumption bundles		
		A	B	C
Prices	A	20	10	22
	B	21	20	15
	C	12	15	10

- * Does WARP hold or is it violated?



How to check the Strong Axiom of Revealed Preference (SARP)

- * 3rd step: Use transitivity to look for indirect revelation of preferences – marked with (*)

Tabela de despesa		Cabazes		
		A	B	C
Preços	A	20	10*	22(*)
	B	21	20	15*
	C	12	15	10

$$A \succ B \wedge B \succ C \Rightarrow A \succ C$$

- * 4th step: Look for situations where both cells (i,j) and (j,i) have an asterisk (with or without brackets).
- * In the table above we find no such situation. Both axioms hold.



Index numbers

- * Index numbers summarize the evolution of a set of variables measured in the same unit.

$$I^{b|t} = \frac{\sum_i w_i x_i^t}{\sum_i w_i x_i^b}$$

Diagram illustrating the components of the index number formula:

- Reference period (points to the superscript t in the numerator)
- Base period (points to the superscript b in the denominator)
- Weight (points to the w_i in the denominator)
- Variable (points to the x_i in the denominator)

Example for 2 variables

$$I^{b|t} = \frac{w_1 x_1^t + w_2 x_2^t}{w_1 x_1^b + w_2 x_2^b}$$



Index numbers (example)

- * In the base year a builder used 100,000 bricks and 80 tons of concrete. $(x_1^b = 100.000 \wedge x_2^b = 80)$
- * In the current year the builder used 120,000 bricks and 70 tons of concrete. $(x_1^t = 120.000 \wedge x_2^t = 70)$
- * How did the quantity of materials used by the builder in his activity changed?
- * Both materials are measured in different units: we have to convert the variables to the same unit of measurement using the weights.
- * Ex.: we could use the volume of each material or its price.



Index numbers (example)

- * Let's use the current price, for example.

$$p_1^t = \text{€}0,2 / \text{brick} \wedge p_2^t = \text{€}60 / \text{ton}$$

$$I^{b|t} = \frac{w_1 x_1^t + w_2 x_2^t}{w_1 x_1^b + w_2 x_2^b} = \frac{0,2 \times 120.000 + 60 \times 70}{0,2 \times 100.000 + 60 \times 80} = \frac{28.200}{24.800} \approx 1,137$$

- * The quantity of materials used by this builder (weighted by their price) increased 13,7% between the base year and the current year.



Laspeyres and Paasche quantity and price indexes

- * *Quantity indexes* measure the evolution of quantities weighted by the prices (change in weighted average consumption)
- * *Price indexes* measure the evolution of prices weighted by quantities (measure of inflation)
- * *Laspeyres indexes* use weights for the base period
- * *Paasche indexes* use weights for the period for which we want to calculate the index (t)



Laspeyres and Paasche quantity and price indexes

Index numbers	Quantity indexes	Price indexes
Laspeyres indexes (base period weights)	$L_q^{\delta t} = \frac{p_1^{\delta} x_1^t + p_2^{\delta} x_2^t}{p_1^{\delta} x_1^{\delta} + p_2^{\delta} x_2^{\delta}};$ $L_q^{\delta t} = \frac{\sum_i p_i^{\delta} x_i^t}{\sum_i p_i^{\delta} x_i^{\delta}}$	$L_p^{\delta t} = \frac{p_1^t x_1^{\delta} + p_2^t x_2^{\delta}}{p_1^{\delta} x_1^{\delta} + p_2^{\delta} x_2^{\delta}};$ $L_p^{\delta t} = \frac{\sum_i p_i^t x_i^{\delta}}{\sum_i p_i^{\delta} x_i^{\delta}}$
Paasche indexes (weights for the period for which the index is being calculated)	$P_q^{\delta t} = \frac{p_1^t x_1^t + p_2^t x_2^t}{p_1^t x_1^{\delta} + p_2^t x_2^{\delta}};$ $P_q^{\delta t} = \frac{\sum_i p_i^t x_i^t}{\sum_i p_i^t x_i^{\delta}}$	$P_p^{\delta t} = \frac{p_1^t x_1^t + p_2^t x_2^t}{p_1^{\delta} x_1^t + p_2^{\delta} x_2^t};$ $P_p^{\delta t} = \frac{\sum_i p_i^t x_i^t}{\sum_i p_i^{\delta} x_i^t}$



Expenditure index

$$M^{b|t} = \frac{p_1^t x_1^t + p_2^t x_2^t}{p_1^b x_1^b + p_2^b x_2^b} = \frac{m^t}{m^b} = \frac{m^b + \Delta m^{b|t}}{m^b} = 1 + \frac{\Delta m^{b|t}}{m^b}$$

$$M^{b|t} = \frac{\sum_i p_i^t x_i^t}{\sum_i p_i^b x_i^b}$$



Relationship between Laspeyres and Paasche quantity and price indexes, the expenditure index, revealed preferences and welfare

- * Using the principle of revealed preference, we can draw some conclusions about consumers' welfare from the Laspeyres and Paasche quantity and price indexes

Index numbers	Quantity indexes	Price indexes
Laspeyres indexes (base period weights)	$L_q^{\delta t} > 1 \Rightarrow ???$ $L_q^{\delta t} < 1 \Rightarrow ???$	$L_p^{\delta t} > M^{\delta t} \Rightarrow ???$ $L_p^{\delta t} < M^{\delta t} \Rightarrow ???$
Paasche indexes (weights for the period for which the index is being calculated)	$P_q^{\delta t} > 1 \Rightarrow ???$ $P_q^{\delta t} < 1 \Rightarrow ???$	$P_p^{\delta t} > M^{\delta t} \Rightarrow ???$ $P_p^{\delta t} < M^{\delta t} \Rightarrow ???$



Relationship among Laspeyres and Paasche quantity and price indexes, the expenditure index, revealed preferences and welfare

* Example:

$$P_q^{\delta t} > 1 \Leftrightarrow \frac{p_1^t x_1^t + p_2^t x_2^t}{p_1^t x_1^\delta + p_2^t x_2^\delta} > 1 \Leftrightarrow p_1^t x_1^t + p_2^t x_2^t > p_1^t x_1^\delta + p_2^t x_2^\delta \Leftrightarrow m^t > p_1^t x_1^\delta + p_2^t x_2^\delta$$

This means that the consumption bundle chosen in the **base** year was affordable in year t , at those year's prices, but was not chosen, therefore $(x_1^t, x_2^t) \succ (x_1^\delta, x_2^\delta)$, which means that the consumer *is better off*.

- * Try to evaluate the remaining cases and confirm your result in the following table.



Relationship among Laspeyres and Paasche quantity and price indexes, the expenditure index, revealed preferences and welfare

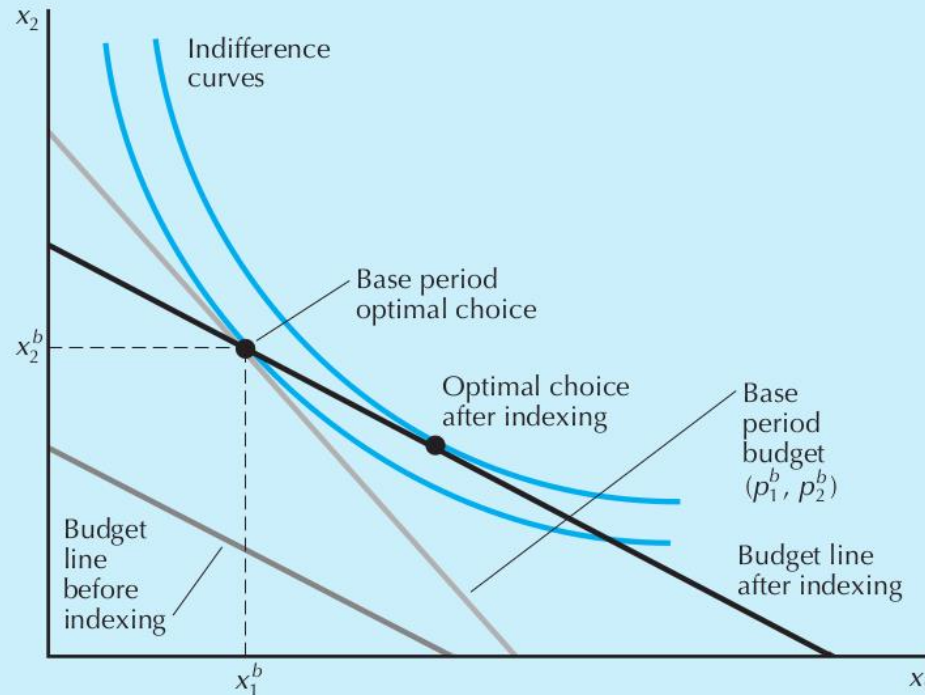
* Results:

Index numbers	Quantity indexes	Price indexes
Laspeyres indexes (base period weights)	$L_q^{\delta t} > 1 \Rightarrow$ no conclusion $L_q^{\delta t} < 1 \Rightarrow$ the consumer is worse off	$L_p^{\delta t} > M^{\delta t} \Rightarrow$ no conclusion $L_p^{\delta t} < M^{\delta t} \Rightarrow$ the consumer is better off
Paasche indexes (weights for the period for which the index is being calculated)	$P_q^{\delta t} > 1 \Rightarrow$ the consumer is better off $P_q^{\delta t} < 1 \Rightarrow$ no conclusion	$P_p^{\delta t} > M^{\delta t} \Rightarrow$ the consumer is worse off $P_p^{\delta t} < M^{\delta t} \Rightarrow$ no conclusion



Laspeyres price index

Revealed
Preferences
and income
indexed to
Laspeyres price
index:



Social Security. Changing prices will typically make the consumer better off than in the base year.

